

On the Intrinsic Dimensional Structure of Transformer Attention

Luis Chamberlain
mcgrof.c@samsung.com

December 5, 2025

LayerNorm removes one dimension; learning determines the rest.

Geometric Constraint

LayerNorm centers each token vector by subtracting its mean component. This eliminates the $[1, 1, \dots, 1]$ direction and places every input to attention on a $(d-1)$ -dimensional hyperplane. All subsequent projections operate entirely within this reduced geometry. The geometry shows only that full rank is unnecessary; it does not predict how much lower the learned effective rank may become.

2D $\rightarrow$ 1D Example

For $\mathbf{x} = [x_1, x_2]$ with mean $\mu = (x_1 + x_2)/2$,

$$\mathbf{z} = [x_1 - \mu, x_2 - \mu]$$

satisfies

$$z_1 + z_2 = 0 \quad \Rightarrow \quad z_2 = -z_1.$$

Every centered vector lies on the line spanned by $[1, -1]$. Stacking N such vectors produces a rank-1 matrix:

$$\begin{bmatrix} 1.2 & -1.2 \\ -0.8 & 0.8 \\ 2.0 & -2.0 \\ 0.5 & -0.5 \end{bmatrix} = \begin{bmatrix} 1.2 \\ -0.8 \\ 2.0 \\ 0.5 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix}.$$

Generalization to d Dimensions

LayerNorm enforces

$$z_1 + z_2 + \dots + z_d = 0,$$

so all token embeddings lie in a $(d-1)$ -dimensional subspace. Thus any matrix of centered embeddings satisfies

$$\text{rank}(Z) \leq d-1.$$

Since attention uses linear projections,

$$Q = ZW_q, \quad K = ZW_k,$$

it follows that

$$\text{rank}(Q), \text{rank}(K) \leq d-1.$$

All similarity structure in $QK^\top$ arises from these $(d-1)$ available directions.

Learned Dimensionality

In practice, trained attention mechanisms often rely on a much smaller subset of directions inside the LayerNorm hyperplane. The architectural constraint provides the ambient $(d-1)$ -dimensional space; learning determines how many of those dimensions are actually used. Thus the intrinsic geometry defines the ceiling, while the effective rank emerges from optimization and data.